BUSINESS SCHOOL

SKEMA

Corporate Finance
PGE M1 Fall 2026

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Session 7: Risk and Return

Housekeeping

About me

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Plan For Today

- Revisiting Solaria Energia
- Risk and Return Statistics
- Some Facts
- Portfolio Mean and Variance
- Appendix: The Efficient Frontier

Computing Growth Rates

For an accounting variable x_{t+1} measured at $t+1$,

$$\text{growth rate of } x_{t+1} = \frac{x_{t+1} - x_t}{x_t}$$

If $\text{Sales}_{2025} = 100$ and $\text{Sales}_{2026} = 125$, we say revenues grew by 25% in 2026.

$$\text{Sales Growth}_{2026} = \frac{125 - 100}{100} = 25\%$$

This is NOT about expectations for 2027.

Computing Stock Returns

Given the price P_{t+1} and dividends paid D_{t+1} at $t+1$, the stock return from t to $t+1$ is computed as

$$
\begin{aligned}
r_{t+1} &= \frac{D_{t+1} + P_{t+1} - P_t}{P_t} \\
&= \underbrace{\frac{D_{t+1}}{P_t}}_{\text{div. yield}} + \underbrace{\frac{P_{t+1} - P_t}{P_t}}_{\text{capital gains}}
\end{aligned}
$$

This is the net return, a number like 5%, in contrast to the *gross* return, $R_{t+1} = 1 + r_{t+1} = 1.05$

Computing Stock Returns



If asked to compute the stock return over a fiscal year using financial statements, the simple way to do this is:

$$r_{t+1} = \frac{\sum_{q=1}^{4} D_q + P_{t+1} - P_t}{P_t}$$

where $D_{t+1} = \sum_{q=1}^{4} D_q$ is the sum of the quarterly dividends paid throughout the year.

This assumes:

- Dividends are not re-invested
- Prices are not already adjusted for dividends.

Computing Stock Returns (Technicalities)



Dividends can be re-invested into the stock at the time of payment.

Then, annual gross returns based on quarterly dividends are

$$R_{t+1} \doteq 1 + r_{t+1} = \prod_{q=1}^{4}(1 + r_q) = (1 + r_1)(1 + r_2)(1 + r_3)(1 + r_4)$$

where

$$1 + r_q = \frac{P_q + D_q}{P_{q-1}}$$

is the quarterly return, computing using beginning and end-of-quarter prices (usually end of March, June, Sep, Dec)

Computing Stock Returns (Technicalities)



Assume there are no dividends, or that prices are dividend-adjusted.

Then,

$$R_{t+1} = \prod_{q=1}^{4}(1 + r_q) = \frac{P_1}{P_0}\frac{P_2}{P_1}\frac{P_3}{P_2}\frac{P_4}{P_3} = \frac{P_4}{P_0}$$

Then, the annual gross return is just dividing Dec 31_{2026} by Dec 31_{2025} prices (for example).

Most data vendors give you the “adjusted close” price which allow you to just divide prices without worrying about dividends.

Free Float and EPS

Shares outstanding in the 10-K is the total number of shares outstanding.
Free float is the subset of outstanding shares that are available for public trading.

They exclude

- Shares held by insiders (CEOs)
- Strategic or controlling holders
- Restricted stock / equity compensation for employees

For most large public companies, you can treat this number as $> 95\%$.
On the other hand, ARM, which IPO'ed in 2023, floats less than 10% of its stock.

Free Float and EPS



EPS applies to all shareholders, regardless of how much stock it floats.
EPS calculations get messy because

- Exercise of stock options by employees
  - Even if not exercised, they may count towards EPS if the option is ITM.
- Conversion into equity by convertible-bond holders
- Stock issuance / repurchase by firms
- Vesting of restricted stock

So in practice, dividing Net Income by shares outstanding may not give you the same EPS as what is stated in the IS.

Liquidity



In Finance, there are 100 definitions of liquidity. For this class, we will think of liquidity as

Liquidity. how easily an asset can be bought or sold without too heavily changing the market price.

All else equal, a company with a higher float has more liquid stock (as there is more supply).

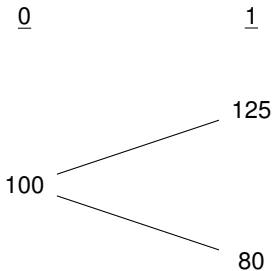
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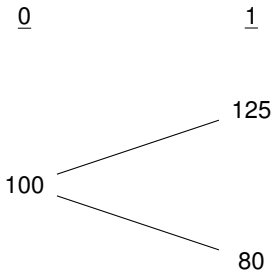
When probabilities are known

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



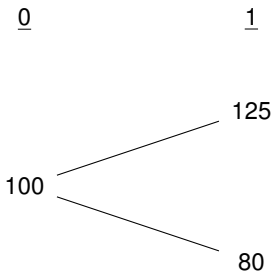
a. If there is equal probability of either state being realized, what is the expected value of the stock in one year?
   102.5

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



b. What is the return on this stock at each state?
25% in the good state, -20% in the bad state.

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



c. What is the expected return on this stock over the next year?
2.5%



Variance is a measure of how spread out outcomes are around the expected value, i.e., the average squared surprise.

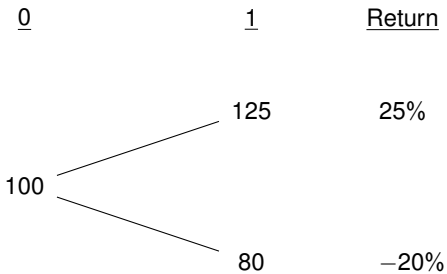
$$\text{Var}(R) = E\left[(R - \mathbb{E}[R])^2\right] = \sum_i \mathrm{p}_i \, (R_i - \mathbb{E}[R])^2$$

where p_i is the probability of each event and $\mathbb{E}[R]$ is the expected value of R.

In our two-state world, the variance is

$$\mathrm{p}_{\text{good}} \times (R_{\text{good}} - \mathbb{E}[R]) + \mathrm{p}_{\text{bad}} \times (R_{\text{bad}} - \mathbb{E}[R])$$

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



d. Calculate the variance of returns.
5.0625%



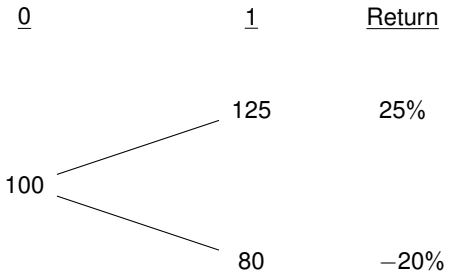
In finance, we define volatility as the standard deviation of a stock's returns, the square-root of the variance,

$$\sigma = \sqrt{\text{Var}(R)} = \sqrt{\sum_i p_i \left(R_i - E[R]\right)^2}$$

Conversely, we write the variance of returns as σ^2.

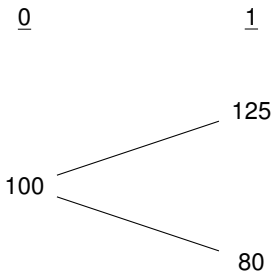
Volatility tells you how much the stock price/return moves around the mean.

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



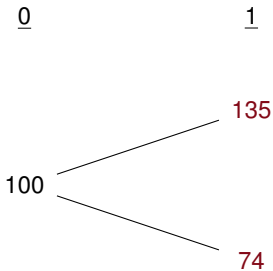
e. Calculate the volatility of returns (Recall, $\sigma^2 = .050625$).
22.5%

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



f. Suppose the probability of the good state is equal to 1. Confirm that the variance is equal to 0.

Suppose we have a non-dividend paying stock that has a price $S_0 = 100$ today. Furthermore, suppose that one year later, the stock will only take either one of two values.



g. Confirm that in the new figure above, where the probability of each event equally likely, the volatility is $\sigma = 30.5\%$.

Exercise 1

The following table shows the one-year return distribution of Startup, Inc.

Probability	40%	20%	20%	10%	10%
Return	-100%	-75%	-50%	-25%	1000%

a. What is the expected return? 32.5%

b. What is the standard deviation of the return? 323.5%

Using historical data to estimate in-sample



Suppose we have the following table of past annual returns for a stock.

Year	Returns
2026	13%
2025	−4%
2024	6%
2023	5%

The average return over some fixed horizon is the simple average,

$$\bar{R} = \frac{1}{T}(R_1 + R_2 + \cdots + R_T) = \frac{1}{T}\sum_{t=1}^{T} R_t$$

Suppose we have the following table of past annual returns for a stock.

Year	Returns
2026	13%
2025	−4%
2024	6%
2023	5%

The volatility, or sample standard deviation of realized returns, measures how much a stock's returns have fluctuated in the past.

$$\bar{\sigma} = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(R_t - \bar{R})^2}$$

The population standard deviation of realized returns is given by

$$\bar{\sigma} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(R_t - \bar{R})^2}$$

*This formula is identical to the one we saw before, except now we are treating the probability of each realization as being equally representative of the distribution.

*The $T - 1$ in the sample standard deviation is a degree-of-freedom correction common in statistics/econometrics since we only observe a sample of the underlying data generating process.

Exercise 2

The last four years of returns for a stock are as follows:

1	2	3	4
-4%	$+28\%$	$+12\%$	$+4\%$

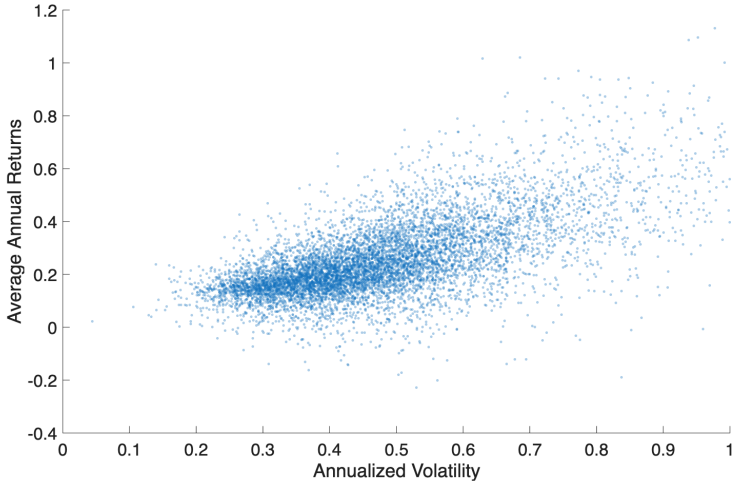
a. What is the average annual return? 10%

b. What is the standard deviation & variance of the stock's returns? $\sigma = 13.66\%$

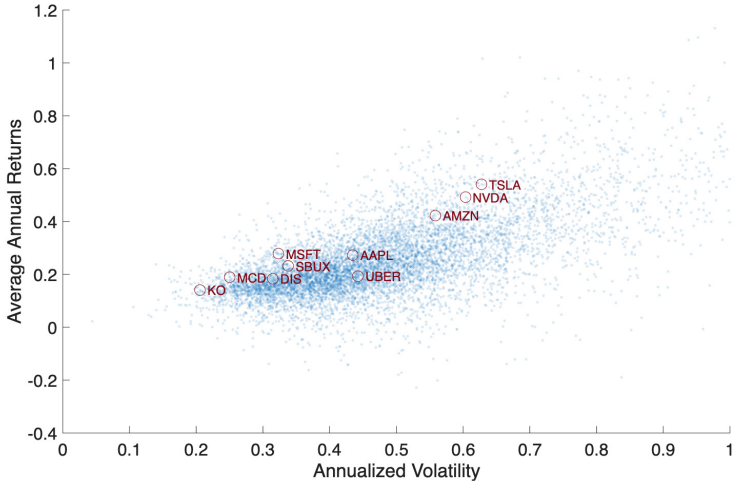
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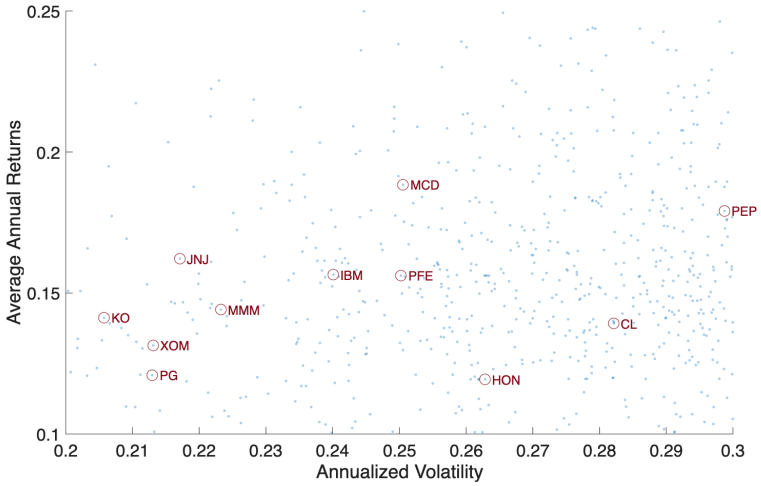
Universe of Stocks: 1926-2025



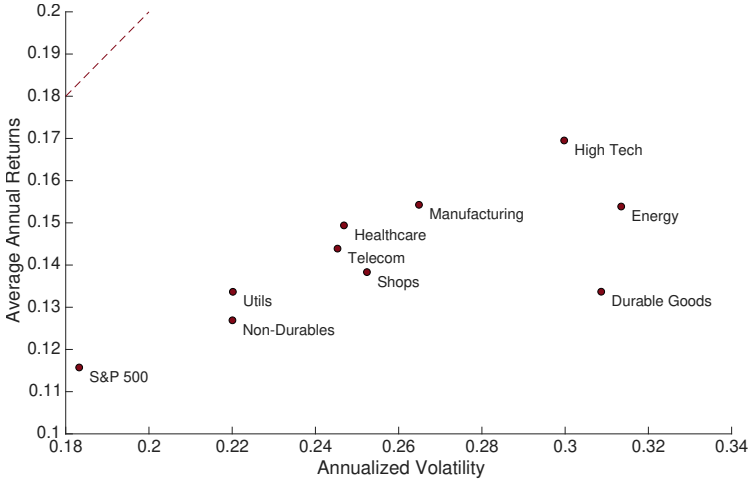
Universe of Stocks: 1926-2025



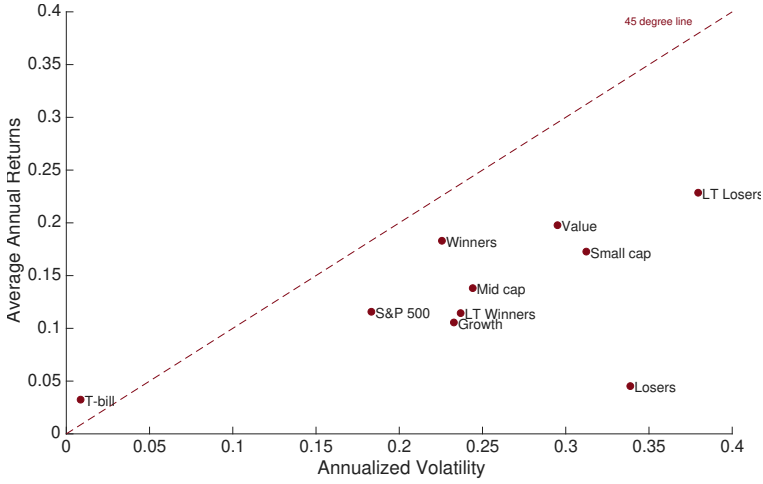
Universe of Stocks: 1926-2025



Industry Portfolios: 1926-2025



Modern Portfolio Choice in 2026



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Portfolio Returns

Suppose we estimate the following in the data:

Stock	Avg Returns	Avg Vol
KO	9%	15%
AMZN	30%	30%

If we invest 40% of our wealth in KO and the remaining 60% in AMZN, what is the expected return of this portfolio?

Portfolio Returns

Suppose we estimate the following in the data:

Stock	Avg Returns	Avg Vol
KO	9%	15%
AMZN	30%	30%

If we invest 40% of our wealth in KO and the remaining 60% in AMZN, what is the expected return of this portfolio?

$$\begin{aligned}\text{Returns}_{\text{portfolio}} &= w \times \text{Returns}_{\text{KO}} + (1-w) \times \text{Returns}_{\text{AMZN}} \\ &= 0.40 \times .09 + 0.60 \times .30 \\ &= 0.036 + 0.18 = 21.6\%\end{aligned}$$

where w is the weight in KO.

Portfolio Returns

For the moment we will assume that expected returns = average historical returns, though we will revise this assumption in the next class.

The expected returns of a portfolio is given by

$$\mathbb{E}[R_p] = \sum_i w_i \times \mathbb{E}[R_i] \quad \text{such that} \quad \sum_i w_i = 1$$

$\mathbb{E}[R_p]$: expected return of the portfolio

$\mathbb{E}[R_i]$: expected return on asset i

w_i: portfolio weight invested in asset i (as a fraction)

Covariance between Two Assets



Recall, volatility measures how much a stock's returns deviate from its average. When we invest in a portfolio of many assets, we want to know whether one asset does poorly when another asset does poorly, and vice-versa.

Covariance measures how two random variables move together relative to their own averages.

$$\text{cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

and when estimated in sample with T observations,

$$\text{cov}(X, Y) = \frac{1}{T-1} \sum_{t=1}^{T} (X_t - \mathbb{E}[X])(Y_t - \mathbb{E}[Y])$$

Covariance between Two Assets

Suppose we have the following table of past annual returns for KO and AMZN. Assume the expected returns on KO is 9% and the expected returns on AMZN is 30%.

Year	KO	AMZN
2026	25%	10%
2025	15%	5%
2024	9%	40%
2023	−5%	80%

Calculate the sample covariance between KO and AMZN stock returns between 2023-2026.

Sample Covariance

Suppose we have the following table of past annual returns for KO and AMZN. Assume the expected return on KO is 9% and the expected return on AMZN is 30%.

Year	KO	AMZN	KO-$\mathbb{E}$[KO]	AMZN-$\mathbb{E}$[AMZN]	Product
2026	0.25	0.10	0.16	-0.20	-0.0320
2025	0.15	0.05	0.06	-0.25	-0.0150
2024	0.09	0.40	0.00	0.10	0.0000
2023	-0.05	0.80	-0.14	0.50	-0.0700
				Sum of products	-0.1170
				Covariance $= -0.1184/3$	-0.0390

Correlation

The correlation between two random variables normalizes the covariance to a number between -1 and 1. It tells you how strongly the two variables move together,

$$\text{Correlation} = \rho = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

- $\rho = +1$: A perfect positive linear relationship
- $\rho = 0$: No linear relationship
- $\rho = -1$: A perfect negative linear relationship.

Correlation

We found that $\text{cov}(\text{KO}, \text{AMZN}) = -.0390$. Assume $\sigma_{\text{KO}} = 15\%$ and $\sigma_{\text{AMZN}} = 30\%$. What is the correlation between KO and AMZN?

$$\text{Correlation} = \rho = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Plugging in the numbers,

$$\rho = \frac{-0.0390}{.15 \times .30} = -0.87$$

In the recent data, KO and AMZN tend move in opposite directions.

Portfolio Variance of Two Risky Assets



Let *w* be the weight invested in KO and (1-w) be the weight invested in AMZN. The portfolio variance is given by

$$\text{var}(p) = \sigma_p^2 = w^2\sigma_{\text{KO}}^2 + (1-w)^2\sigma_{\text{AMZN}}^2 + 2w(1-w)\text{cov}(\text{KO}, \text{AMZN})$$

which is equal to

$$\text{var}(p) = \sigma_p^2 = w^2\sigma_{\text{KO}}^2 + (1-w)^2\sigma_{\text{AMZN}}^2 + 2w(1-w)\rho\sigma_{\text{KO}}\sigma_{\text{AMZN}}$$

*From

$$\text{var}(aX + bY) = a^2\text{var}(X) + b^2\text{var}(Y) + 2ab\text{cov}(X, Y)$$

Portfolio Variance of Two Risky Assets



If we invest 40% of our wealth in KO and the remaining 60% in AMZN, what is the volatility of this portfolio?
Assume $\rho = -0.87$ and $\sigma_{\text{KO}} = 15\%$ and $\sigma_{\text{AMZN}} = 30\%$.

$$\text{var}(p) = \sigma_p^2 = w^2\sigma_{\text{KO}}^2 + (1-w)^2\sigma_{\text{AMZN}}^2 + 2w(1-w)\rho\sigma_{\text{KO}}\sigma_{\text{AMZN}}$$

Portfolio Variance of Two Risky Assets



If we invest 40% of our wealth in KO and the remaining 60% in AMZN, what is the volatility of this portfolio?
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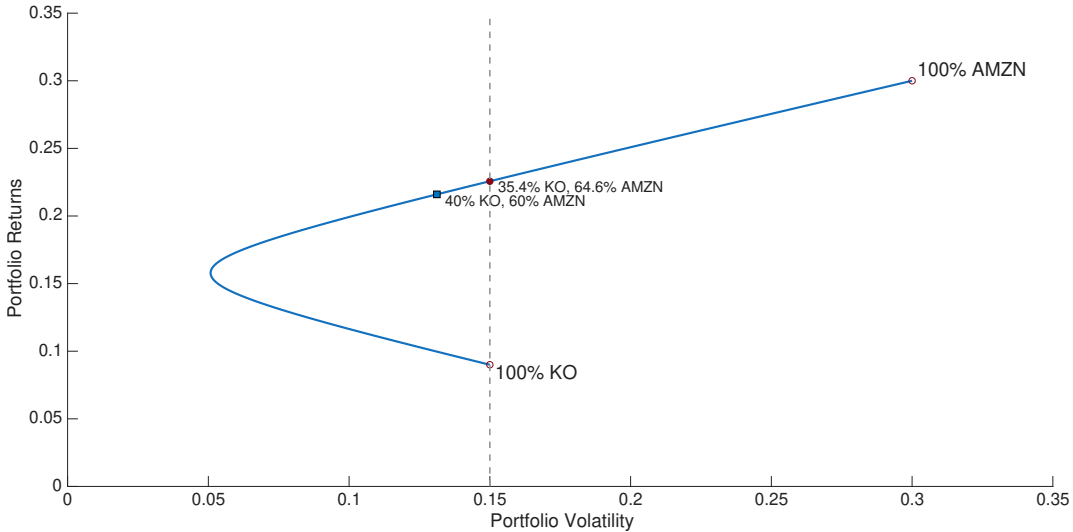
$$\text{var}(p) = \sigma_p^2 = w^2\sigma_{\text{KO}}^2 + (1-w)^2\sigma_{\text{AMZN}}^2 + 2w(1-w)\rho\sigma_{\text{KO}}\sigma_{\text{AMZN}}$$

Plugging in,

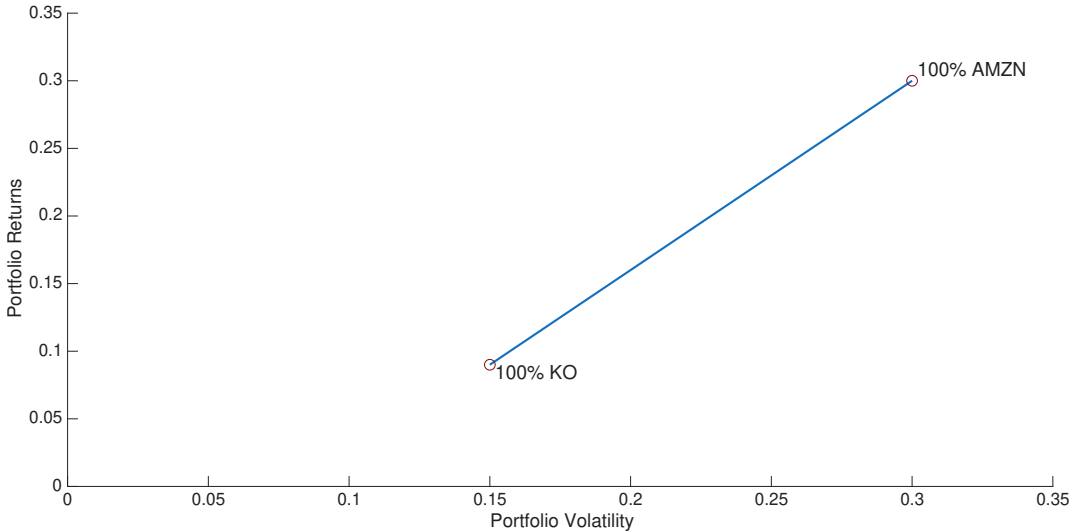
$$\sigma_p^2 = 0.4^2(0.15)^2 + 0.6^2(0.30)^2 + 2(0.4)(0.6)(-0.87)(0.15)(0.30) = 0.0172$$
$$\sigma_p = \sqrt{0.0172} \approx 13.1\%$$

Recall, the expected returns on this portfolio was 21.6%. Due to the strong negative correlation between KO and AMZN, our portfolio volatility is lower than the volatility of KO despite having more than twice the expected returns.

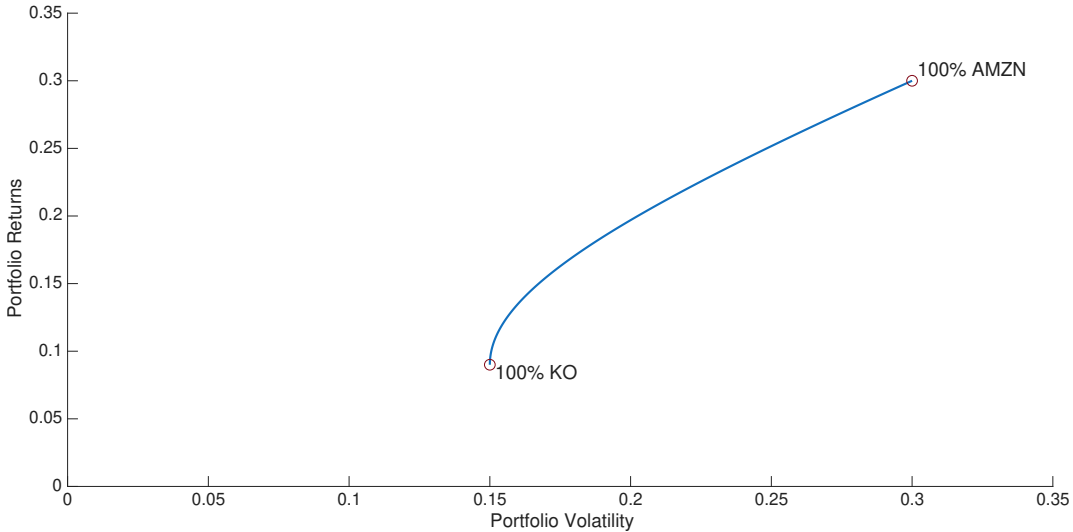
Risk-Return Plane with Two Risky Assets, $\rho = -0.87$



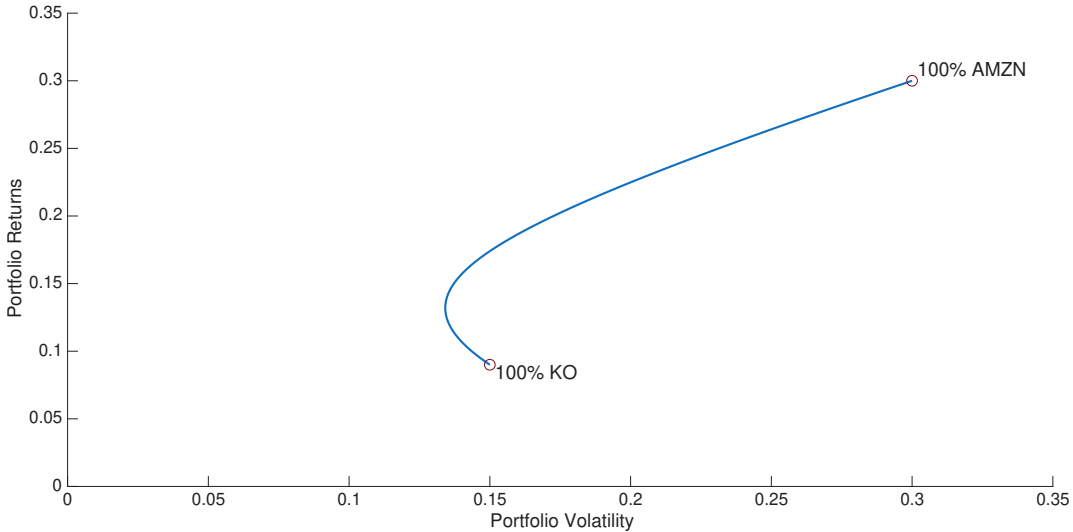
Risk-Return Plane with Two Risky Assets, $\rho = 1$



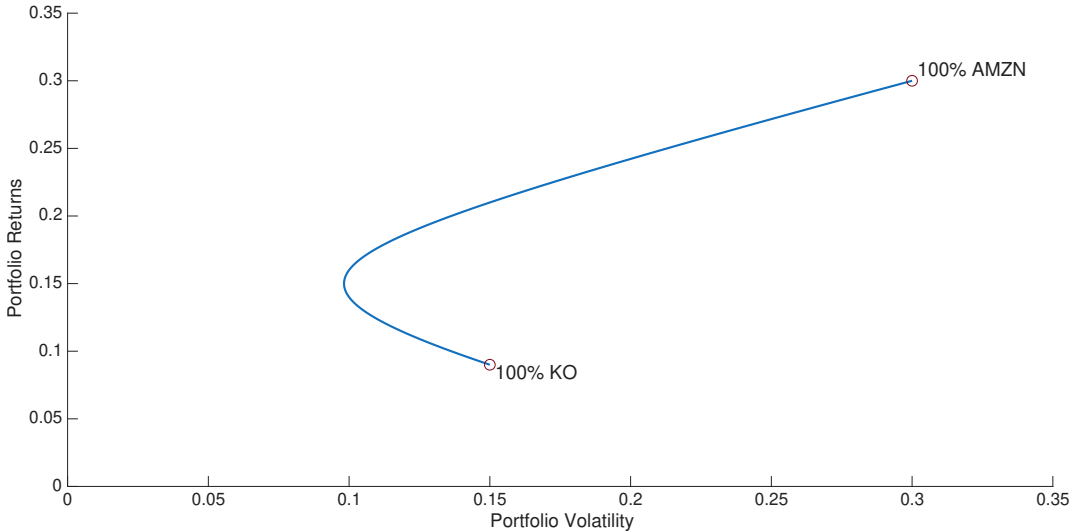
Risk-Return Plane with Two Risky Assets, $\rho = 0.5$



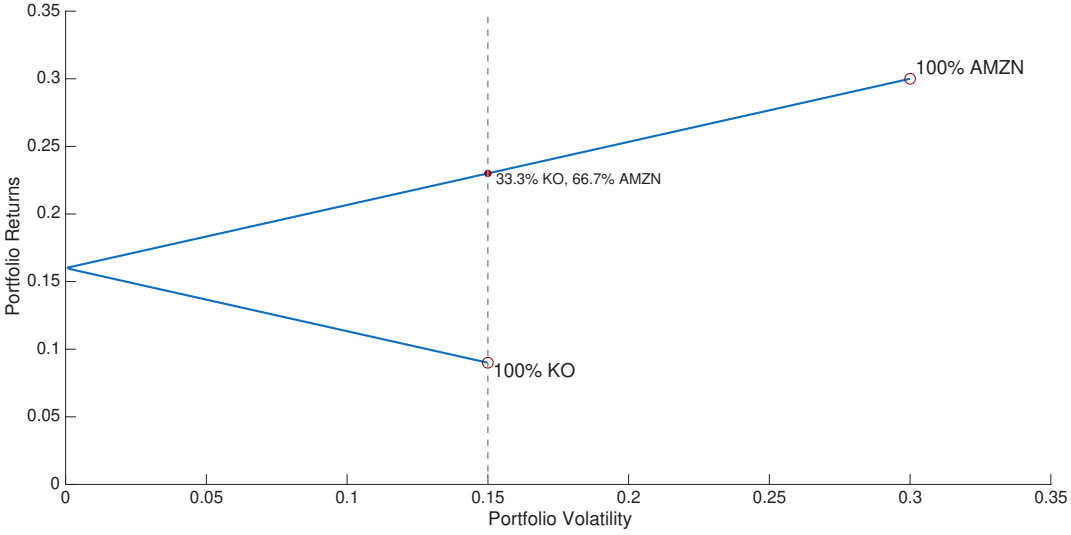
Risk-Return Plane with Two Risky Assets, $\rho = 0$



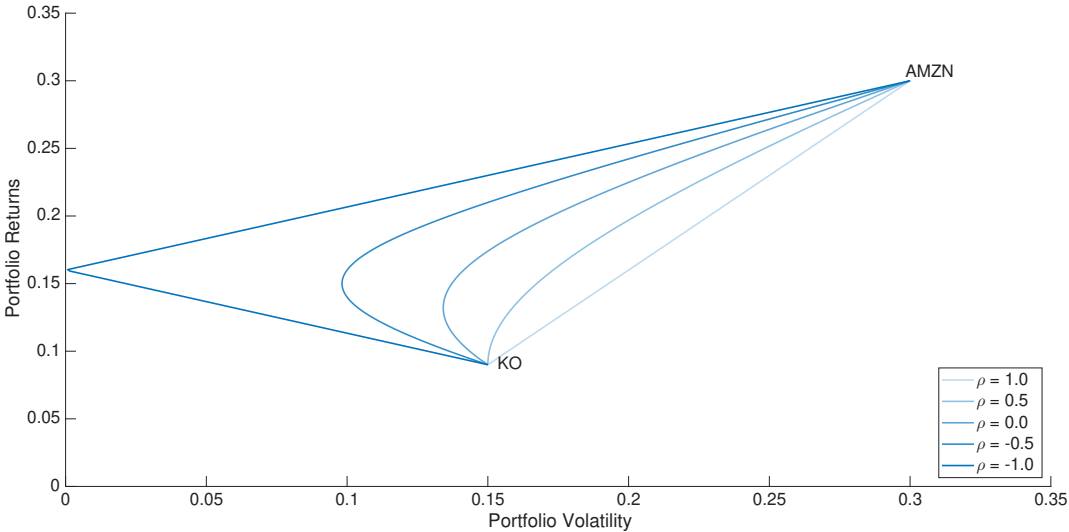
Risk-Return Plane with Two Risky Assets, $\rho = -0.5$



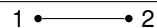
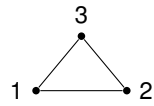
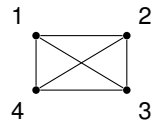
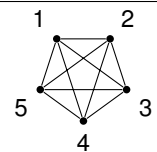
Risk-Return Plane with Two Risky Assets, $\rho = -1$



Risk-Return Plane with Two Risky Assets



Portfolio Variance with Many Assets

2 Assets		1 distinct **correlation**
3 Assets		3 distinct **correlations**
4 Assets		6 distinct **correlations**
5 Assets		10 distinct **correlations**

For N assets, there are $N(N-1)/2$ correlation pairs, so correlations entirely determine the portfolio variance when N is large.

Limit Results on Portfolio Diversification

Recall,

$$\sigma_p^2 = \text{var}(aX + bY) = a^2\text{var}(X) + b^2\text{var}(Y) + 2ab\text{cov}(X, Y)$$

More generally for N assets,

$$\sigma_p^2 = \sum_i \sum_j w_i w_j \sigma_{ij}$$

for $i = 1...N$ and $j = 1..N$, where σ_{ij} is the covariance between asset i and asset j.

When $i = j$, it is just the variance of the asset.

Limit Results on Portfolio Diversification

Assume an equal-weighted portfolio, $w_i = w_j = 1/N$,

$$\sigma_p^2 = \frac{1}{N^2} \sum_i \sum_j \sigma_{ij}$$

for $i = 1...N$ and $j = 1..N$.

There are N times that $i = j$, so there are N variance terms.
There are $N(N-1)$ times that $i \neq j$, so there are $N(N-1)$ covariance terms.

Limit Results on Portfolio Diversification

Splitting into a variance component and a covariance component,

$$\sigma_p^2 = \frac{1}{N^2}\left(\sum_{i=j}\sigma_{ij} + \sum_{i\neq j}\sigma_{ij}\right)$$

Defining $\bar{\sigma}^2 \doteq \frac{1}{N}\sum_{i=j}\sigma_{ij}$ as the average variance,
and $\overline{\text{cov}} \doteq \frac{1}{N(N-1)}\sum_{i\neq j}\sigma_{ij}$ as the average covariance,

$$\begin{aligned}\sigma_p^2 &= \frac{1}{N^2}\left(N\bar{\sigma}^2 + N(N-1)\overline{\text{cov}}\right)\\ &= \left(\frac{1}{N}\right)\bar{\sigma}^2 + \left(\frac{N-1}{N}\right)\overline{\text{cov}}\end{aligned}$$

Limit Results on Portfolio Diversification

Our portfolio variance is

$$\sigma_p^2 = \left(\frac{1}{N}\right)\bar{\sigma}^2 + \left(\frac{N-1}{N}\right)\overline{\text{cov}}$$

Taking limits as $N \to \infty$

$$\lim_{N\to\infty} \sigma_p^2 = \overline{\text{cov}}$$

Our portfolio variance is determined by the average covariance between all assets as N becomes large.

Appendix: Limit Results on Portfolio Diversification

In matrix notation, let $\mathbf{w}$ be a $Nx1$ vector of portfolio weights, and Σ a NxN variance-covariance matrix. Then, portfolio variance with equal weights is

$$\sigma_p^2 = \mathbf{w}'\Sigma\mathbf{w} = \frac{1}{N^2}\mathbf{1}'\Sigma\mathbf{1}$$

Decompose Σ into the diagonal and off-diagonal components as before,

$$\mathbf{1}'\Sigma\mathbf{1} = \mathbf{1}'\mathbf{V}\mathbf{1} + \mathbf{1}'\mathbf{C}\mathbf{1}$$

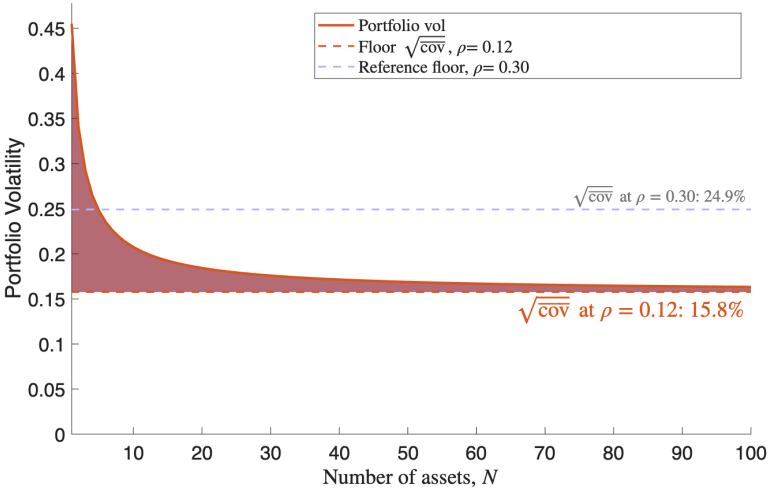
where $\mathbf{V} \doteq \text{diag}(\sigma_1^2, ..., \sigma_N^2)$ and $\mathbf{C} = \Sigma - \mathbf{V}$ is a NxN matrix of off-diagonal components. Then,

$$\sigma_p^2 = \frac{1}{N^2}\left(\text{tr}(\Sigma) + \mathbf{1}'\mathbf{C}\mathbf{1}\right)$$

where $\text{tr}(\Sigma)$ is the trace of a matrix (the sum of the diagonals), and using $\bar{\sigma}^2 \doteq \text{tr}(\Sigma)/N$ and $\overline{\text{cov}} = (\mathbf{1}'\Sigma\mathbf{1} - \text{tr}(\Sigma))/(N(N-1))$,

$$\sigma_p^2 = \left(\frac{1}{N}\right)\bar{\sigma}^2 + \left(\frac{N-1}{N}\right)\overline{\text{cov}}$$

Portfolio Volatility: $\bar{\sigma} = 0.45,\ \bar{\rho} = 0.12$



Market Risk vs Idiosyncratic Risk



There are many names, but firm-specific, idiosyncratic, or diversifiable risk refers to good or bad news about the company itself.

Example. The risk of a CEO targeted by a foreign intelligence officer in an extramarital affair.

An investor, by holding a diversified portfolio of many stocks, can reduce his/her portfolio volatility to $\sqrt{\overline{\text{cov}}}$. This implies that if an event contributes nothing to $\sqrt{\overline{\text{cov}}}$, the "price" of this event is equal to zero.

A Return Decomposition

For stock i, let

$$r_{t+1}^i = \mu^i + \sqrt{\vartheta^i} m + \sqrt{1 - \vartheta^i} \epsilon^i$$

The return is decomposed into

- The average stock return μ^i
- A common market component m that affects all firms (unpredictable)
- A firm-specific component ϵ^i (unpredictable)

With $\vartheta^i \in [0, 1]$ as a normalization, as the loading on the market risk.
Moreover, assume m and ϵ^i have mean zero and unit variance and
$\text{cov}(m, \epsilon^i) = 0$
$\text{cov}(\epsilon^i, \epsilon^j) = 0$
Then,

$$\text{cov}(r_{t+1}^i, r_{t+1}^j) = \sqrt{\vartheta^i \vartheta^j}$$

A Return Decomposition

Assuming unit variance for all stocks,

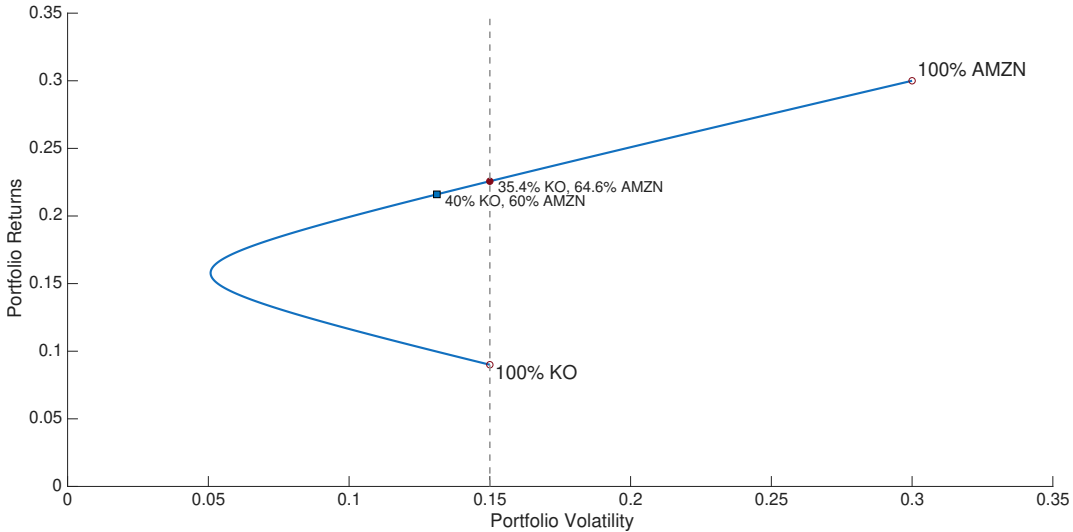
$$\operatorname{cov}(r^i_{t+1}, r^j_{t+1}) = \sqrt{\vartheta^i \vartheta^j}$$

By assumption, firm-specific shocks ϵ^i do not contribute to covariance as $\operatorname{cov}(\epsilon^i, \epsilon^j) = 0$ and $\operatorname{cov}(m, \epsilon^i) = 0$.
From before, $\overline{\text{cov}} \approx \sqrt{\bar{\vartheta}}$, where $\bar{\vartheta}$ is the average loading on the market component across all stocks.

Plan For Today

- Revisiting Solaria Energia
- Risk and Return Statistics
- Some Facts
- Portfolio Mean and Variance
- Appendix: The Efficient Frontier

Risk-Return Plane with Two Risky Assets, $\rho = -0.87$



The Efficient Frontier

KO had an average return of 9% and volatility of $\sigma_{\text{KO}} = 15\%$. A portfolio of 35.4% in KO and 64.6% in AMZN gives
Portfolio returns

$$\begin{aligned}\mathbb{E}[R_p] &= w\mathbb{E}[R_{\text{KO}}] + (1-w)\mathbb{E}[R_{\text{AMZN}}] \\ &= .354 \times .09 + .646 \times .3 \\ \mathbb{E}[R_p] &= 22.6\%\end{aligned}$$

Portfolio volatility,

$$\sigma_p^2 = w^2\sigma_{\text{KO}}^2 + (1-w)^2\sigma_{\text{AMZN}}^2 + 2w(1-w)\rho\sigma_{\text{KO}}\sigma_{\text{AMZN}}$$
$$\sigma_p^2 = 0.354^2(0.15)^2 + 0.646^2(0.30)^2 + 2(0.354)(0.646)(-0.87)(0.15)(0.30) \approx 0.0225$$
$$\sigma_p = \sqrt{0.0225} = 15\%$$

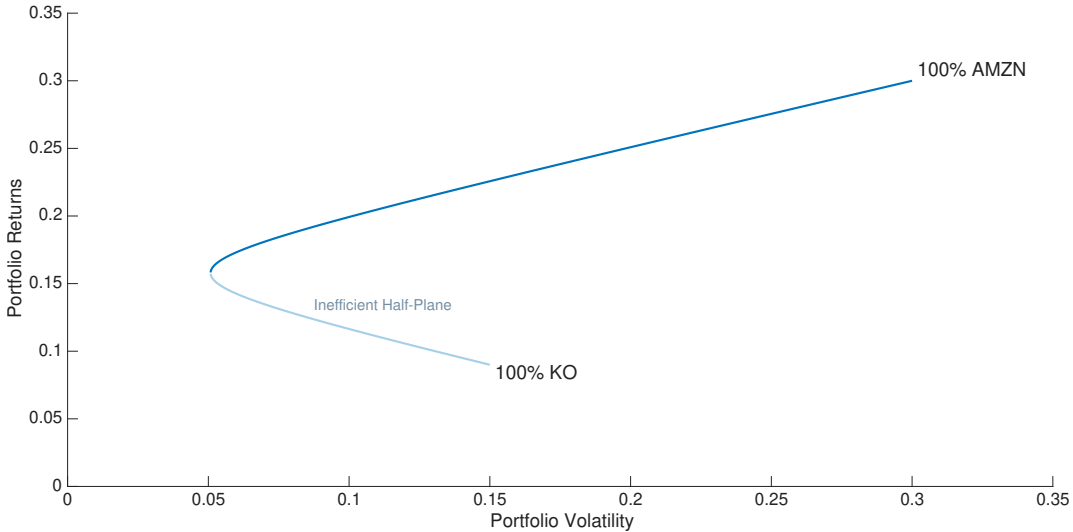
The Efficient Frontier



KO had an average return of 9% and volatility of $\sigma_{\text{KO}} = 15\%$. A portfolio of 35.4% in KO and 64.6% in AMZN gives expected returns of 22.6% for the same level of volatility of 15%.

Hence, we say that investing in KO is dominated by a combination of KO and AMZN, and that it is inefficient to invest 100% in KO.

The Efficient Frontier



The Efficient Frontier



An inefficient portfolio is a portfolio in which it is possible to find another portfolio (using the same set of assets) that gives a higher expected return for the same level of risk.

A stock may be useful regardless of its own mean and variance due to its correlation to other stocks.

The Efficient Frontier with Many Assets



See the file “Frontier Builder - Week 7 (Optional).html”, along with “stock data - Week 7 (Optional).csv” on K2.

Sharpe Ratio

The Sharpe Ratio of a portfolio is defined as

$$\text{Sharpe Ratio}_p = \frac{R_p - r^f}{\sigma_p}$$

It measures how much return you earn above the risk-free interest rate per unit of risk (volatility) in your portfolio.

The tangent portfolio is the portfolio with the highest Sharpe ratio in the efficient frontier of all assets.

The Market Portfolio



If we assume (among many other things) that all investors have identical beliefs about means, variances, and covariances across all assets, then everyone agrees about the composition of the tangent portfolio.

The tangent portfolio is the best investment in the investor's opportunity set. If everyone wants the same portfolio, by supply and demand, the composition of the market portfolio will exactly equal the composition of the tangent portfolio.

If every investor wants to hold 5% of their portfolio in Apple, the price of Apple stock will go up due to investor demand until it represents exactly 5% of the wealth in the economy.

What if I’m more risk-averse than the average investor?



If we only invest in stocks but we want less volatility in our portfolio, we will have to trade not just the mean, but also the Sharpe ratio of the portfolio (because the tangent portfolio has the highest).

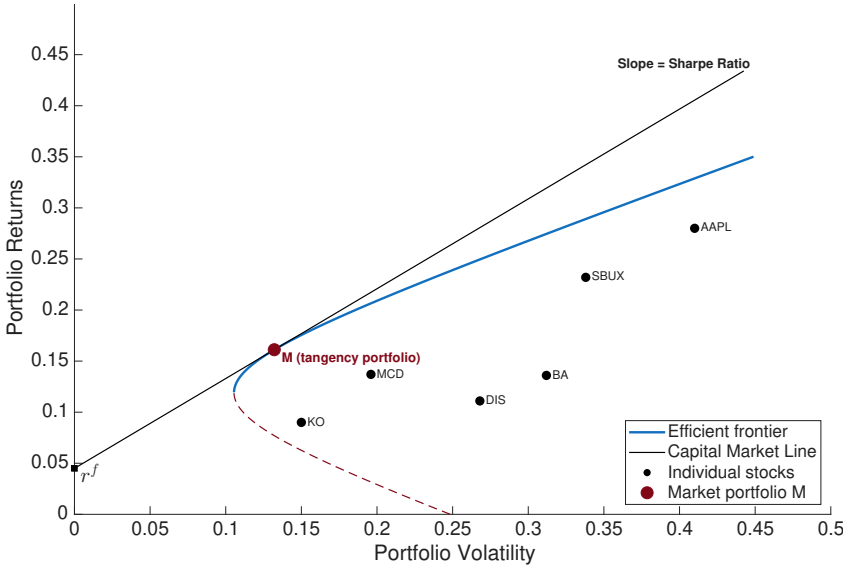
What if I'm more risk-averse than the average investor?



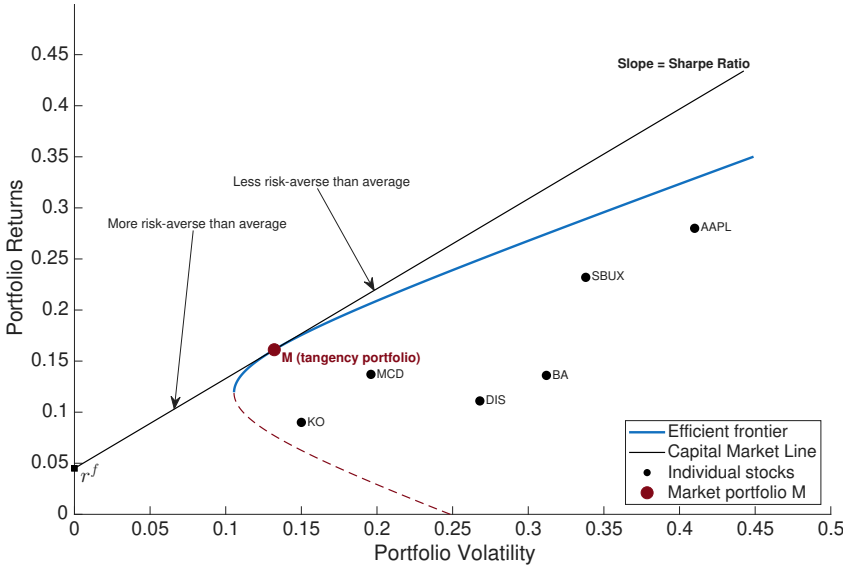
Instead of investing in a portfolio with a lower Sharpe ratio, we can invest in a portfolio of the risk-free asset (a government bond) and the market portfolio.

Two-Fund Separation Theorem. There exists two portfolios, the risk-free asset and the market portfolio, such that the optimal portfolio of any investor is a combination of these two assets.

The Capital Market Line



The Capital Market Line



Derivation of the CML

Consider a portfolio with two assets r^f and the market portfolio R^m. Assume the weight invested in the market portfolio is w. Then,

$$\mathbb{E}[R_p] = (1 - w)r^f + w\mathbb{E}[R^m]$$

Factoring, we get an equivalent representation,

$$\mathbb{E}[R_p] = r^f + w(\mathbb{E}[R^m] - r^f)$$

Find the variance of this portfolio. Assuming the variance on the risk-free asset is equal to zero (because it is assumed to be riskless), and the covariance between the risk-free asset and the market portfolio to be zero (it is not true in practice), we find that

$$\sigma_p = w\sigma^m$$

where σ^m is the volatility of the market portfolio. Using $w = \frac{\sigma_p}{\sigma^m}$,

$$\mathbb{E}[R_p] = r^f + \frac{\mathbb{E}[R^m] - r^f}{\sigma^m} \times \sigma_p$$

Derivation of the CML

The Capital Market Line is given by the equation

$$\mathbb{E}[R_p] = r^f + \frac{\mathbb{E}[R^m] - r^f}{\sigma^m} \times \sigma_p$$

where the slope is the Sharpe ratio on the market/tangent portfolio. To apply the CML, we either start with a target portfolio return or a target portfolio volatility, then identify the weights in each asset using

$$\mathbb{E}[R_p] = (1 - w)r^f + w\mathbb{E}[R^m]$$

based on the expected portfolio returns given the target portfolio vol or target portfolio returns.

Summary



From a theoretical point of view (with many strong assumptions), the two-fund separation theorem says that simply investing in the market portfolio gives you a better risk-adjusted return than trying to "beat-the-market" through picking individual stocks.

In practice, researchers have found zero evidence of mutual fund managers' ability to pick stocks. In fact, the average performance is slightly negative due to fees.

The share of "passive" investing versus "active" investing has been steadily increasing in the past 15 years, but individual stock picking is still needed for informational efficiency.